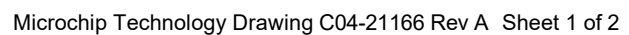
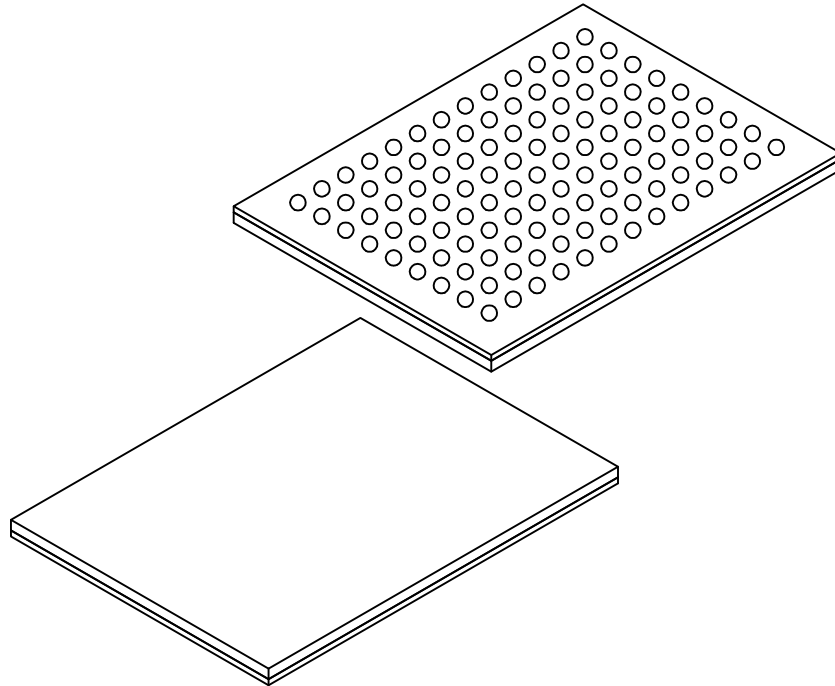


Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



117-Ball Ultra Thin Fine-Pitch Ball Grid Array Package (C5B) - 9.5x7.0x0.65 mm Body [UFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



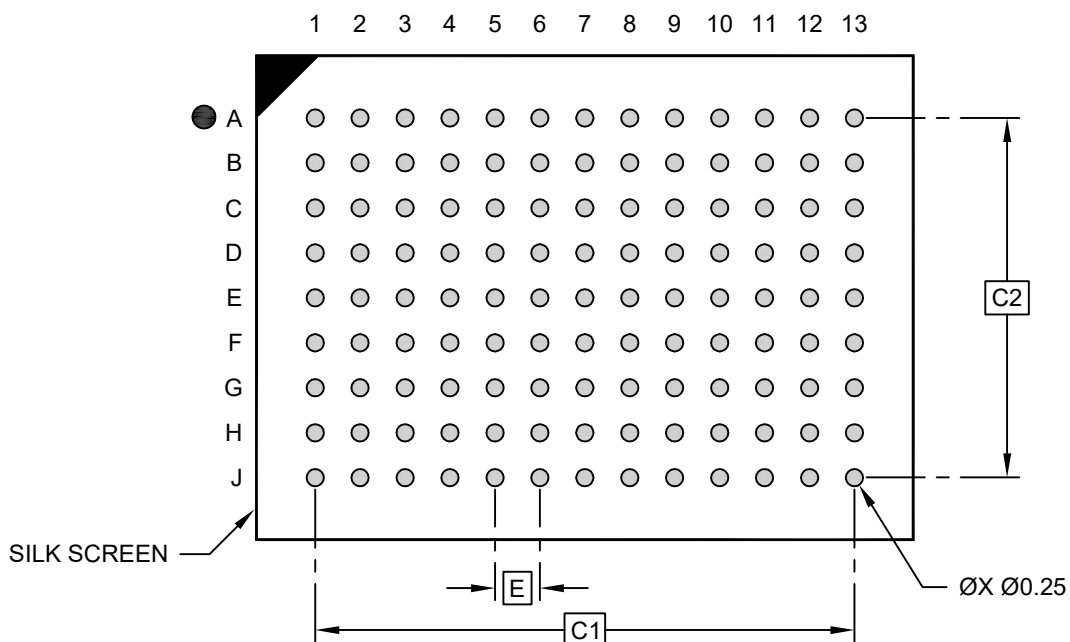
Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	117		
Pitch	e	0.65 BSC		
Overall Height	A	–	–	0.65
Ball Height	A1	0.16	0.21	0.26
Mold Thickness	M	0.25 REF		
Substrate Thickness	S	0.136 REF		
Overall Length	D	9.50 BSC		
Ball Array Length	D1	7.80 BSC		
Overall Width	E	7.00 BSC		
Ball Array Width	E1	5.20 BSC		
Ball Diameter	b	0.25	0.30	0.35

Notes:

- Ball A1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

117-Ball Ultra Thin Fine-Pitch Ball Grid Array Package (C5B) - 9.5x7.0x0.65 mm Body [UFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Contact Pad Spacing	C1	7.80 BSC		
Contact Pad Spacing	C2	5.20 BSC		
Contact Pad Width (Xnn)	X			0.25

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23166 Rev A